Ankitha Betala

Propulsion CFD Engineer
Aerospace Simulation Engineer — Thermal-Fluids

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Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2023–2025	M.Tech (Aero Thermal)	Indian Institute of Technology, Kanpur	6.5/10
2023	B.E. (Aeronautical Engineering)	Hindusthan Institute of Technology, Coimbatore	8.86/10
2019	Board of Intermediate Education(XII)	Sri Chaitanya Jr. Mahila Kalasala, Vijayawada	9.92/10
2017	Board of Secondary Education(X)	Andhra Pradesh Residential School, Musunuru	9.5/10

Technical Skills

- **CFD & Fluid Mechanics:** ANSYS Fluent, Compressible Flow, RANS Modelling, SST k–ω Turbulence Model, Shock-Wave Analysis, Internal Flow, Nozzle Flow, Exhaust-Plume Interaction
- **Propulsion:** Solid Rocket Motors, Internal Ballistics, Propulsion Performance Analysis, NASA CEA
- **Thermal Sciences:** Conjugate Heat Transfer (CHT), Convective Heat Transfer, Thermal Management, Temperature and Heat-Flux Analysis
- **Multiphysics & Structural:** ANSYS Mechanical, Static Structural Analysis, Linear and Nonlinear Analysis, Fluid–Structure Interaction (FSI)
- **Programming & Data Analysis:** MATLAB, Simulink, Python, C, LaTeX, NI LabVIEW
- **HPC & Computing:** High Performance Computing (HPC), Linux Environment, Remote Cluster Computing, MobaXterm
- **Tools:** ANSYS Workbench

Professional Experience

Engineer II (*Jan'26–Present*)
INVENTGRID INDIA PRIVATE LIMITED (IG Defence)
Department: New Development – CFD Analysis

- **JWALA Defence Project**
  - Performed compressible-flow CFD simulations of propulsion nozzle configurations using ANSYS Fluent to investigate internal flow development, pressure and velocity distributions, and exhaust-plume behaviour.
  - Applied the RANS-based SST k–ω turbulence model to resolve near-wall flow behaviour and investigate shock structures and plume development.
  - Post-processed axial profiles and flow-field contours, with the analysed results showing local Mach numbers approaching 3.9 and static-pressure levels of approximately 4.7 MPa in the plotted axial profile.
  - Evaluated velocity and temperature fields, with contour ranges extending to approximately 1.82 km/s and 1600 K, respectively, to interpret high-speed exhaust-flow characteristics.
  - Conducted conjugate heat transfer (CHT) and thermal assessments to investigate coupled fluid-flow and solid heat-conduction behaviour and identify thermally critical regions.
  - Performed fluid–structure interaction (FSI) analysis to assess the effect of flow-induced pressure loading on structural response.
  - Interpreted CFD and thermal results and prepared engineering reports to support propulsion-system evaluation and design discussions.
- **KAL RATO Booster Project**
  - Post-processed static test data from D1 and D3 motor tests to extract thrust-time and chamber-pressure histories and evaluate propulsion performance for a 250 kg-class RATO system.
  - Calculated required thrust and impulse for rail inclinations of 15°–20° and target release speeds of 160–190 km/h.
  - Evaluated first-three-second impulse values of 14,571.429 N s (D1) and 14,339.812 N s (D3) against the nominal requirement of 14,040.189 N s.
  - Compared measured pressure and thrust histories to assess motor response during the useful initial three-second operating period.
  - Performed compressible-flow CFD analysis of rocket motor chamber and nozzle configurations using ANSYS Fluent to investigate flow development and exhaust-plume characteristics.
  - Conducted conjugate heat transfer (CHT) analysis to investigate coupled gas-flow and solid thermal behaviour, including temperature distribution and heat-transfer characteristics in propulsion components.
  - Performed structural and FSI assessments to evaluate pressure-induced deformation and the interaction between fluid loading and component response.

Research Interests

- High-Speed Compressible Flows, Shock-Wave Physics and Propulsion CFD
- Gas Turbine Aerodynamics, Thermal Management and Heat Transfer
- Conjugate Heat Transfer and High-Temperature Flow Physics
- Hydrogen Combustion and Turbulent Reacting Flows
- Combustion Instabilities, Thermoacoustics and Acoustic–Flame Interaction
- Coherent Flow Structures, Modal Analysis and Data-Driven Flow Analysis

Key Projects

M.Tech Thesis: Conjugate Heat Transfer Analysis of a Circular Nozzle with Passive Cooling (*Aug'2024–2025*)

Proposed by: Aeronautical Development Establishment (ADE), DRDO

Supervisor: Dr. Abhijith Kushari — Indian Institute of Technology Kanpur

- Conducted numerical investigation of conjugate heat transfer in a circular nozzle with passive cooling using **ANSYS Fluent**.
- Investigated coupled reacting-flow, convection and solid heat-conduction behaviour to evaluate nozzle thermal performance.
- Analysed temperature distribution, wall heat flux, cooling effectiveness and thermal gradients to identify thermally critical regions.
- Evaluated ejector-assisted passive cooling behaviour and its influence on nozzle thermal management.
- Performed simulations on High Performance Computing (HPC) clusters using Linux-based remote computing environments.

Signal Processing and Coherent Flow Structure Analysis Using MATLAB

Independent Computational Studies — Measurement and Data Analysis

- Developed MATLAB workflows using auto-correlation, cross-correlation and FFT to investigate periodic behaviour, dominant frequency components and time-delay relationships in synthetic pressure signals.
- Implemented Proper Orthogonal Decomposition (POD) and Spectral Proper Orthogonal Decomposition (SPOD) to identify energetic spatial modes and frequency-dependent coherent structures in synthetic two-dimensional velocity fields.
- Evaluated modal energy distribution, cumulative energy and SPOD spectra; the generated examples showed a leading POD mode with approximately 67% energy and a dominant SPOD spectral peak near 78.12 Hz.

Welding of Dissimilar Aluminium Alloys (Al7075–Al6063) (*Jan'23–May'23*)

Queen India Engineering Services, Chennai

- Investigated TIG welding of dissimilar aluminium alloys using ER4043 filler material.
- Evaluated mechanical properties through hardness, impact and microstructural analysis.
- Performed non-destructive testing (NDT) to identify surface and internal defects.

Course Projects

- **Temperature Measurement of Wind-Driven Pool Fire Using LabVIEW** (*Apr'2024*)
  Guide: Dr. Kamal Poddar — Department of Aerospace Engineering, IIT Kanpur
  - Developed a LabVIEW interface for temperature acquisition and monitoring in pool-fire experiments.
  - Performed data acquisition, processing and temperature measurement analysis.
- **Transient Thermal Analysis of a Sphere with an Enclosure** (*Jan'2024–May'2024*)
  Guide: Dr. Mohammad Ibrahim Sugarno — Department of Aerospace Engineering, IIT Kanpur
  - Conducted transient thermal analysis using ANSYS Workbench.
  - Investigated temperature evolution and heat-transfer characteristics of a solid sphere enclosed by an air domain.
- **Finite Element Analysis Convergence and Mesh Independence** | Certificate

Internships

Solid Rocket Motor Thrust Vector Control (*Oct'22–2023*)

Defence Research and Development Organisation (DRDO)

Advanced Systems Laboratory, Hyderabad

- Studied solid rocket motor dynamics, thrust-vectoring principles and nozzle pressure variation.
- Designed an elliptical convergent–divergent nozzle in AutoCAD and subdivided the geometry into three sections.
- Studied thrust-vectoring mechanisms and their role in rocket trajectory control and stability.

Relevant Courses

Heat Transfer	Measurements and Data Analysis
Fundamentals of Combustion	Combustion Diagnostics
Computational Fluid Dynamics	Numerical Methods
Convective Heat Transfer	Rockets and Missiles
Problem Solving and Python Programming	Introduction to Virtual Instrumentation
Finite Element Methods	Experimental Stress Analysis

Positions of Responsibility

- **Teaching Assistant, Engineering Graphics (Indian Institute of Technology, Kanpur)** (*Aug'2024–Jul'2025*)
  - Prepared written and computer-based assignments for undergraduate students.
  - Mentored 30 undergraduate students and reviewed their academic progress.
- **Volunteer, ACGT Conference** (*Aug'2024*)
  - Guided 80 participants during the conference.

Scholastic Achievements

- Secured **All India Rank 1017** in **GATE 2023**.
- Received the **Prathibha Puraskar Award** on SSC merit from the Government of Andhra Pradesh (2017).
- Received the **Ugam Legrand Scholarship** on intermediate merit basis (2019–2023).